

WEEK – 3 BUSINESS ANALYTICS ASSIGNMENT

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

Task Area: Business Analytics (BA)

Objective: To design an optimized supply-chain network connecting factories, warehouses, and customer zones while minimizing total operating expenditure.

1. Introduction

Supply chain network design is the process of deciding how products should move from factories to warehouses and then to customer zones. In a multi-echelon network, decisions must consider production capacity, warehouse capacity, transportation limits, customer demand, and minimum service-level requirements. Mixed-Integer Linear Programming (MILP) is useful because it can optimize both continuous quantities and yes/no decisions such as whether a warehouse should be opened.

2. Problem Statement

A company operates several factories and warehouses and serves multiple customer zones. The company must determine the quantity shipped from each factory to each warehouse and from each warehouse to each customer zone. Some warehouses may be opened or closed. The objective is to satisfy customer demand and service-level requirements while respecting production, storage, and transportation capacities and minimizing the total supply-chain cost.

3. Network Structure

Factories → Warehouses → Customer Zones

Factory 1 ■■■■

Factory 2 ■■■■■■ → Warehouse 1 ■■■■

Factory 3 ■■■■ Warehouse 2 ■■■■■■ → Customer Zone 1

Warehouse 3 ■■■■■■ → Customer Zone 2

■■■■ → Customer Zone 3

4. Sets and Parameters

Symbol	Meaning
i	Factory index
j	Warehouse index
k	Customer-zone index
$CapF_{i\bullet}$	Production capacity of factory i
$CapW_{j\bullet}$	Storage capacity of warehouse j
$Dem_{k\bullet}$	Demand of customer zone k
$c_{i\bullet}$	Unit transportation cost from factory i to warehouse j
$d_{j\bullet}$	Unit transportation cost from warehouse j to customer k
$F_{j\bullet}$	Fixed cost for opening warehouse j
$SL_{k\bullet}$	Minimum service level required by customer zone k

5. Decision Variables

- x_{ij} = quantity shipped from factory i to warehouse j .
- y_{jk} = quantity shipped from warehouse j to customer zone k .
- z_j = binary variable: 1 if warehouse j is opened, otherwise 0.

6. Objective Function

The objective is to minimize the total operating expenditure consisting of factory-to-warehouse transportation cost, warehouse-to-customer transportation cost, and fixed warehouse-opening cost.

$$\begin{aligned} \text{Minimize } Z = & \sum(i,j) c[i,j] x[i,j] \\ & + \sum(j,k) d[j,k] y[j,k] \\ & + \sum(j) F[j] z[j] \end{aligned}$$

7. MILP Constraints

7.1 Factory Production Capacity

The total quantity shipped by each factory cannot exceed its production capacity.

$$\sum(j) x[i,j] \leq \text{CapF}[i] \quad \text{for every factory } i$$

7.2 Warehouse Flow Balance

The quantity entering a warehouse must equal the quantity leaving it.

$$\sum(i) x[i,j] = \sum(k) y[j,k] \quad \text{for every warehouse } j$$

7.3 Warehouse Storage Capacity

A warehouse can handle products only when it is opened, and its flow cannot exceed its capacity.

$$\sum(k) y[j,k] \leq \text{CapW}[j] \times z[j] \quad \text{for every warehouse } j$$

7.4 Customer Demand

Each customer zone must receive enough product to satisfy its demand.

$$\sum(j) y[j,k] \geq \text{Dem}[k] \quad \text{for every customer zone } k$$

7.5 Minimum Service Level

The model can enforce a minimum percentage of demand that must be served.

$$\sum(j) y[j,k] \geq \text{SL}[k] \times \text{Dem}[k] \quad \text{for every customer zone } k$$

7.6 Binary Warehouse Decision

$$z[j] \in \{0,1\}$$

All shipment quantities are non-negative: $x[i,j] \geq 0$ and $y[j,k] \geq 0$.

8. Sample Input Data

Factory capacities

Factory	Capacity
F1	500 units
F2	600 units
F3	400 units

Warehouse data

Warehouse	Capacity	Fixed Cost
W1	700 units	■5,000
W2	500 units	■4,000
W3	600 units	■4,500

Customer-zone data

Customer Zone	Demand	Minimum Service Level
C1	400 units	90%
C2	350 units	90%
C3	300 units	90%

9. Python Implementation using PuLP

The following sample program creates the MILP model and solves it using the PuLP optimization library.

```
from pulp import *

factories = ["F1", "F2", "F3"]
warehouses = ["W1", "W2", "W3"]
customers = ["C1", "C2", "C3"]

factory_cap = {"F1": 500, "F2": 600, "F3": 400}
warehouse_cap = {"W1": 700, "W2": 500, "W3": 600}
demand = {"C1": 400, "C2": 350, "C3": 300}

fixed = {"W1": 5000, "W2": 4000, "W3": 4500}

# Example unit costs
cost_fw = {(i,j): 5 for i in factories for j in warehouses}
cost_wc = {(j,k): 7 for j in warehouses for k in customers}

model = LpProblem("Supply_Chain_MILP", LpMinimize)

x = LpVariable.dicts("FactoryToWarehouse",
                    [(i,j) for i in factories for j in warehouses],
                    lowBound=0)
y = LpVariable.dicts("WarehouseToCustomer",
                    [(j,k) for j in warehouses for k in customers],
                    lowBound=0)
z = LpVariable.dicts("OpenWarehouse", warehouses,
                    cat="Binary")

model += (
    lpSum(cost_fw[i,j] * x[i,j] for i in factories for j in warehouses)
    + lpSum(cost_wc[j,k] * y[j,k] for j in warehouses for k in customers)
    + lpSum(fixed[j] * z[j] for j in warehouses)
)

for i in factories:
    model += lpSum(x[i,j] for j in warehouses) <= factory_cap[i]

for j in warehouses:
    model += lpSum(x[i,j] for i in factories) == lpSum(y[j,k] for k in customers)
    model += lpSum(y[j,k] for k in customers) <= warehouse_cap[j] * z[j]

for k in customers:
```

```

model += lpSum(y[j,k] for j in warehouses) >= demand[k]
model += lpSum(y[j,k] for j in warehouses) >= 0.90 * demand[k]

model.solve()

print("Status:", LpStatus[model.status])
print("Minimum Cost:", value(model.objective))

for v in model.variables():
    if v.varValue and v.varValue > 0:
        print(v.name, "=", v.varValue)

```

10. Expected Optimization Output

After solving, the program returns the optimal warehouse-opening decisions, shipment quantities, and minimum total cost. For example, the output may show which warehouses are opened and how many units are transported through each route. The exact optimal values depend on the transportation-cost matrix and other input data.

11. Business Interpretation

The MILP model helps management make cost-effective network decisions. If opening a warehouse reduces transportation expenses enough to justify its fixed cost, the model can select that warehouse. If a warehouse is expensive or unnecessary, the binary variable can keep it closed. The model also ensures that factory capacities, warehouse capacities, customer demand, and service levels are respected.

12. Advantages of the MILP Approach

- Minimizes total supply-chain expenditure.
- Supports warehouse opening and closing decisions.
- Considers multiple supply-chain stages simultaneously.
- Handles capacity and service-level constraints.
- Provides a quantitative basis for logistics planning.
- Can be extended to real-world global supply-chain networks.

13. Conclusion

Multi-Echelon Supply Chain Network Design using MILP provides a structured way to optimize the movement of goods from factories through warehouses to customer zones. By combining continuous shipment decisions with binary warehouse decisions, the model identifies a feasible and cost-efficient network. This approach can help businesses reduce logistics expenditure while maintaining capacity and customer service requirements.

End of Assignment